Lecture 7: Collisional hydrodynamics

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1 Basic equations

In this lecture, we develop plasma hydrodynamics using the kinetic equation *with* collisions,

$$\frac{\partial f}{\partial t} + \nabla \cdot (\boldsymbol{v} f) + \nabla_{\boldsymbol{v}} \cdot \left[\frac{q}{m} \left(\boldsymbol{E} + \frac{\boldsymbol{v}}{c} \times \boldsymbol{B} \right) f \right] = C. \tag{1}$$

The fluid equations are derived as usual. Assuming there is no ionization and recombination, collisions conserve particles, so the continuity equation remains the same. The momentum equation and the pressure equation acquire additional terms

$$\boldsymbol{R} = \int m \boldsymbol{v} C \, \mathrm{d}\boldsymbol{v} = \int m \boldsymbol{w} C \, \mathrm{d}\boldsymbol{v}, \qquad \boldsymbol{\mathcal{S}} = \int m \boldsymbol{w}\boldsymbol{w} C \, \mathrm{d}\boldsymbol{v}. \tag{2}$$

This leads to

$$\partial_t n + \nabla \cdot (n \boldsymbol{U}) = 0, \tag{3}$$

$$m \left[\partial_t \boldsymbol{U} + (\boldsymbol{U} \cdot \nabla) \boldsymbol{U} \right] = q \left(\boldsymbol{E} + \boldsymbol{U} \times \boldsymbol{B}/c \right) - (\nabla \cdot \boldsymbol{P})/n + \boldsymbol{R}, \tag{4}$$

$$\partial_t P_{ab} + \boldsymbol{U} \cdot \nabla P_{ab} + P_{ab} \nabla \cdot \boldsymbol{U} + P_{bc} \partial_c U_a + P_{ac} \partial_c U_b + \partial_c \mathcal{Q}_{abc} = \epsilon_{ade} \Omega_e P_{bd} + \epsilon_{bde} \Omega_e P_{ad} + \mathcal{S}_{ab}. \tag{5}$$

These equations have the same problem as collisionless fluid equations that we studied in the previous lecture: they are not a closed system, so we need to develop a closure to solve them. Having weak collisions does not significantly change the closure problem. However, if collisions are *strong enough*, then the inverse collision rate can be used as an additional small parameter and new closures can be constructed. Depending on the assumed ordering and types of collisions, such closures can take various forms. Here, we will overview some common ones and outline a general procedure for how other collisional closures can be constructed.

2 Crude collisional closure: weakly ionized plasma

First, let us consider strong collisions in weakly ionized plasmas, because they allow for a particularly simple treatment. In this case, collisions can be modeled using the BGK operator

$$C_s = -\nu_s (f_s - f_{s0}), \tag{6}$$

where ν_s is a constant and f_{s0} is a local Maxwellian with density $\int f_s \, \mathrm{d}\boldsymbol{v}$. We assume that charged particles collide mainly with neutrals, and those are heavy enough to be considered

motionless. Provided that the collision rate ν_s is sufficiently large, one can expect ions to be mostly in a local thermodynamical equilibrium with neutrals. This means that $\boldsymbol{P}_i$ is diagonal and the ion temperature remains close to the temperature T_n of the ambient neutral gas, so ions can be considered isothermal. For electrons, energy exchange with neutrals is suppressed by a factor m_e/m_n, so it takes longer for T_e to approach T_n than for T_i to approach T_n. Still, with enough time, such relaxation still can occur, and we will assume that it does, because this leads to a particularly instructive model to study.

2.1 Density equation

Since we already worked with the isothermal closure before, one might ask: what else is here to study? Well, previously, we did not have collisions, and we used $T = \text{const}$ as a replacement of the equation for $\partial_t \boldsymbol{P}$. Now, with strong enough collisions, we do not even have to solve for $\partial_t \boldsymbol{U}$. This is seen as follows. Under the assumption of (6), the friction force is

$$\boldsymbol{R}_s = -\int m_s \boldsymbol{w} \nu_s (f_s - f_{s0})\,\mathrm{d}\boldsymbol{v} = -m_s \nu_s \int \boldsymbol{v}(f_s - f_{s0})\,\mathrm{d}\boldsymbol{v} = -m_s \nu_s n_s \boldsymbol{U}_s. \tag{7}$$

Assuming that the dynamics of interest is slow, meaning $\partial_t \ll \nu_s$ and $\boldsymbol{U}_s \cdot \nabla \ll \nu_s$, we can neglect the left-hand side in (4). Then, (4) can be reduced to the following equation:

$$\underbrace{\nu_s \boldsymbol{U}_s - \boldsymbol{U}_s \times \boldsymbol{\Omega}_s}_{\boldsymbol{N}_s \boldsymbol{U}_s} \approx \frac{q_s}{m_s}\boldsymbol{E} - v_{Ts}^2 \nabla \ln n_s, \tag{8}$$

where $\boldsymbol{N}_s$ is a matrix given by

$$\boldsymbol{N}_s = \begin{pmatrix} \nu_s & -\Omega_s & 0 \\ \Omega_s & \nu_s & 0 \\ 0 & 0 & \nu_s \end{pmatrix}, \qquad \boldsymbol{N}_s^{-1} = \begin{pmatrix} \frac{\nu_s}{\nu_s^2+\Omega_s^2} & \frac{\Omega_s}{\nu_s^2+\Omega_s^2} & 0 \\ -\frac{\Omega_s}{\nu_s^2+\Omega_s^2} & \frac{\nu_s}{\nu_s^2+\Omega_s^2} & 0 \\ 0 & 0 & \frac{1}{\nu_s} \end{pmatrix}. \tag{9}$$

Then, one obtains $\boldsymbol{U}_s = \boldsymbol{\mu}_s \boldsymbol{E} - v_{Ts}^2 \boldsymbol{N}_s^{-1} \nabla \ln n_s$, where $\boldsymbol{\mu}_s = (q_s/m_s)\boldsymbol{N}_s^{-1}$ is called the particle mobility.[1] Correspondingly, the flux density $\boldsymbol{\Gamma}_s \doteq n_s \boldsymbol{U}_s$ is

$$\boldsymbol{\Gamma}_s = \underbrace{\frac{n_s q_s}{m_s}\boldsymbol{N}_s^{-1}}_{\boldsymbol{\sigma}_s/q_s}\boldsymbol{E} - \underbrace{v_{Ts}^2 \boldsymbol{N}_s^{-1}}_{\boldsymbol{D}_s} \nabla n_s. \tag{10}$$

Substituting this into the continuity equation (3), one obtains

$$\partial_t n_s = \nabla \cdot (\boldsymbol{D}_s \nabla n_s - q_s^{-1} \boldsymbol{\sigma}_s \boldsymbol{E}). \tag{11}$$

The part of the flux $\boldsymbol{\Gamma}_s$ that is proportional to the electric field is $\boldsymbol{\sigma}_s \boldsymbol{E}$ gives rise to a conductive current $\boldsymbol{\sigma}_s \boldsymbol{E}$, so $\boldsymbol{\sigma}_s$ can be understood as the conductivity. The terms $\sigma_{\mathrm{P},s} \equiv \sigma_{s,xx} = \sigma_{s,yy}$

[1] Do not confuse the mobility with the magnetic moment, which is also denoted $\boldsymbol{\mu}$. Also note that some authors define the mobility as $\boldsymbol{\mu}_s = (|q_s|/m_s)\boldsymbol{N}_s^{-1}$ instead, i.e. with the opposite sign for electrons.

and $\sigma_{\mathrm{H},s} \equiv \sigma_{s,yx} = -\sigma_{s,yx}$ (so that $\sigma_{\mathrm{H},e} > 0$) are known as the Pedersen conductivity and the Hall conductivity, respectively, and are given by

$$\sigma_{\mathrm{P},s} = \frac{n_s q_s^2}{m_s} \frac{\nu_s}{\nu_s^2 + \Omega_s^2}, \qquad \sigma_{\mathrm{H},s} = -\frac{n_s q_s^2}{m_s} \frac{\Omega_s}{\nu_s^2 + \Omega_s^2}. \tag{12}$$

(At $\nu_s/\Omega_s \ll 1$, the currents $\sigma_{xy}E_y$ and $\sigma_{yx}E_x$ are, clearly, just the $E \times B$ currents.) The other term, $-\boldsymbol{D}_s \nabla n_s$ is called diffusive flux, or Fick's flux,[2] and makes (12) a diffusion equation. Let us discuss this in detail.

2.2 Diffusion coefficients

Was it possible to anticipate that n_s satisfies a diffusion equation? To answer this, let us look into how collisions work, starting with parallel motion. A typical particle moves freely with velocity $\sim v_{Ts}$ along $\boldsymbol{B}$ and thus, between collisions, shifts by a distance of the order of the mean free path $\lambda_{\mathrm{mfp},s} = v_{Ts}/\nu_s$. The collision time ν_s^{-1} is defined as the time scale on which collisions randomize the free motion; i.e., after a collision, the particle can be moving in either direction. Then the particle motion along $\boldsymbol{B}$ can be considered as a series of random jumps such that, on time $\Delta t \sim \nu_s^{-1}$, the typical displacement is such that $\langle (\Delta z)^2 \rangle \sim \lambda_{\mathrm{mfp},s}^2$. This process can be described using the Fokker–Planck equation with the corresponding diffusion coefficient

$$D_{\parallel,s} \sim \lambda_{\mathrm{mfp},s}^2 \nu_s \sim v_{Ts}^2/\nu_s. \tag{13}$$

This coincides with $D_{zz,s}$ that governs the density relaxation along the z-axis in (12).

Now let us consider transverse dynamics. Between collisions, a particle undergoes cyclotron motion at frequency Ω_s. Suppose that the particle transverse velocity changes sign at a collision event. This does not instantaneously change the particle location, but it does place the particle on a different Larmor orbit whose center is shifted by two gyroradii $\rho_s \equiv v_{Ts}/\Omega_s$ away from the center of the original orbit. Assuming $\nu_s \ll \Omega_s$, the particle can be anywhere on that second orbit when it collides next time. This means that a particle randomly jumps by the distance $\sim \rho_s$ every collision time ν_s^{-1} and thus, according to the Fokker–Planck equation, is supposed to diffuse across the magnetic field with the diffusion coefficient

$$D_{\perp,s} \sim \rho_s^2 \nu_s \sim v_{Ts}^2 \nu_s/\Omega_s^2. \tag{14}$$

This coincides with $D_{xx,s}$ and $D_{yy,s}$ (at $\nu_s \ll \Omega_s$) that govern the density relaxation along the x- and y-axis in (12). The coefficients $D_{xy,s} = -D_{yx,s}$ describe diamagnetic flows with velocities

$$\boldsymbol{U}_{\mathrm{D},s} = \frac{\Omega_s^2}{\nu_s^2 + \Omega_s^2} \frac{v_{Ts}^2}{\Omega_s} \boldsymbol{b} \times \nabla \ln n_s = \frac{1}{1 + (\nu_s/\Omega_s)^2} \frac{cT}{q_s} \boldsymbol{b} \times \ln \nabla n_s \tag{15}$$

(remember the discussion about plasma diamagnetism?) and do not actually produce diffusion.

Notice also that Fick's term causes n_s to evolve at rates $L_\parallel^2/D_{\parallel,s} \sim (L_\parallel/\lambda_{\mathrm{mfp},s})^2/\nu_s$ and $L_\perp^2/D_{\perp,s} \sim (L_\perp/\rho_\perp)^2/\nu_s$, as seen from (12) by taking $\nabla_\parallel \sim 1/L_\parallel$ and $\nabla_\perp \sim 1/L_\perp$. Since we assumed $\partial_t \ll \nu_s$, this implies that our formulation holds only when

$$\lambda_{\mathrm{mfp},s} \ll L_\parallel, \qquad \rho_s \ll L_\perp. \tag{16}$$

[2] When $\boldsymbol{\sigma}_s \boldsymbol{E}$ in (10) is negligible, the formula $\boldsymbol{\Gamma}_s = -\boldsymbol{D}_s \nabla n_s$ is called Fick's law.

These assumptions are implicitly hidden in our expression for the friction force $\boldsymbol{R}_s = -m_s\nu_s n_s \boldsymbol{U}_s$ (or, equivalently, our collision operator (6)). In case of nonlocal transport, the friction force cannot be modeled with such a simple formula.

2.3 Ambipolar diffusion

Consider electron–ion plasma with one type of ions. Clearly, the diffusion coefficients for electrons and ions are different. Then, even when the electron and ion density are approximately equal to each other, the corresponding Fick's fluxes will be different. As a result, charge will build up, and $\boldsymbol{E}$ will grow until the system reaches equilibrium. Let us consider how this works.

2.3.1 Nonmagnetized plasma

In nonmagnetized plasma ($\Omega_s = 0$), the fluxes are simply

$$\boldsymbol{\Gamma}_s = n_s\mu_s\boldsymbol{E} - D_s\nabla n_s. \tag{17}$$

Requiring that the total current is zero, $q_e\Gamma_e + q_i\Gamma_i \approx 0$, leads to

$$(q_e n_e)\mu_e\boldsymbol{E} - D_e\nabla(q_e n_e) + (q_i n_i)\mu_i\boldsymbol{E} - D_i\nabla(q_i n_i) \approx 0. \tag{18}$$

Assuming that the plasma is quasineutral, we have $q_e n_e \approx -q_i n_i$. Then, dividing (18) by $q_e n_e$ yields

$$\mu_e\boldsymbol{E} - D_e\nabla\ln n - \mu_i\boldsymbol{E} + D_i\nabla\ln n \approx 0, \tag{19}$$

where n is either of the densities. This leads to

$$\boldsymbol{E} = \frac{D_i - D_e}{\mu_i - \mu_e}\nabla\ln n, \tag{20}$$

which is called the *ambipolar field*. Then, the flux densities are

$$\boldsymbol{\Gamma}_i = \mu_i\frac{D_i - D_e}{\mu_i - \mu_e}\nabla n_i - D_i\nabla n_i = -\frac{\mu_i D_e - \mu_e D_i}{\mu_i - \mu_e}\nabla n_i \tag{21}$$

and $\boldsymbol{\Gamma}_e = Z_i\boldsymbol{\Gamma}_e$. In summary then,

$$\boldsymbol{\Gamma}_s = -D_a\nabla n_s, \qquad D_a = \frac{\mu_i D_e - \mu_e D_i}{\mu_i - \mu_e}, \tag{22}$$

so both species diffuse with the same diffusion coefficient D_a. This is called *ambipolar diffusion*, and D_a is called the ambipolar diffusion coefficient. For collisions with neutrals, the cross section is roughly constant, so the collision rate scales as $\nu_s \propto v_{Ts} \propto m_s^{-1/2}$. This means that $\mu_s = q_s/m_s\nu_s \propto m_s^{-1/2}$, whence $\mu_e \gg \mu_i$. Using this and $D_s = T\mu_s/q_s$, one readily finds that

$$D_a = (1 + Z_i^{-1})\frac{T}{e}\frac{\mu_i|\mu_e|}{\mu_i + |\mu_e|} \approx (Z_i + 1)\frac{T\mu_i}{q_i} = (Z_i + 1)D_i \sim D_i \ll D_e. \tag{23}$$

In other words, electrostatic interaction between electrons and ions accelerates ion diffusion by a factor $Z_i + 1$ and slows down electron diffusion by a large factor $\sim\sqrt{m_i/m_e}$ so that both species would diffuse at the same rate.

2.3.2 *Magnetized plasma*

In magnetized plasma, diffusion is more complicated. Parallel diffusion is stronger for electrons, like in nonmagnetized plasma, but the transverse diffusion is stronger for ions. Then, fluxes do not necessarily need to be ambipolar: ions can diffuse out radially, thus carrying out positive current, while electrons can diffuse out along field lines, thus carrying out negative current. If the system is short-circuited, for example, through a wall, the overall fluxes remain ambipolar (in the sense that the total current leaving the system is zero) but the perpendicular and parallel diffusion do not need to be ambipolar separately. The correct equation to solve in this case is not $\boldsymbol{j} = 0$, as we assumed above, but $\nabla \cdot \boldsymbol{j} = 0$. The solution depends on a specific geometry, there is no simple universal solution in this case.

3 Asymptotic collisional closures

3.1 Introduction

Reducing all fluid equations to just one fluid equation for the density, used in the previous section, has the advantage of leading to a very simple model. Unfortunately, this model is very limited. To describe broader plasma dynamics, one has to consider fluid equations not only for n but also for $\boldsymbol{U}$ and $\boldsymbol{P}$. This is similar to the case of collisionless hydrodynamics. What is different, though, is that now we have a new parameter, the collision frequency ν.

Let us assume that collisions are strong, so that the distribution is close to Maxwellian,[3]

$$f_0(t, \boldsymbol{x}, \boldsymbol{v}) = n \left(\frac{m}{2\pi T}\right)^{3/2} \exp\left(-\frac{m(\boldsymbol{v} - \boldsymbol{U})^2}{2T}\right), \tag{24}$$

where n, $\boldsymbol{U}$, and T can depend on $(t, \boldsymbol{x})$. In this case, the pressure is mostly a scalar, so the traceless part of the pressure tensor and the heat flux can be treated as a perturbation. It then makes sense to abandon the pressure-tensor equation and utilize a simpler equation for the scalar pressure, which is derived as follows.

3.2 A scalar alternative to the pressure-tensor equation

Let us assume the splitting $P_{ab} = P\delta_{ab} + \Pi_{ab}$, where $\boldsymbol{\Pi}$ is the traceless part of the pressure tensor, and consider the evolution of the scalar pressure $P \doteq \operatorname{tr} \boldsymbol{P}/D$, where D is the number of dimensions.[4] By taking the trace of (5), one obtains:

$$D(\partial_t P + \boldsymbol{U} \cdot \nabla P + P \nabla \cdot \boldsymbol{U}) + 2 \operatorname{tr}(\boldsymbol{P} \nabla \boldsymbol{U}) + 2 \nabla \cdot \boldsymbol{Q} = 2\epsilon_{ade} P_{ad} \Omega_e + \operatorname{tr} \boldsymbol{\mathcal{S}}, \tag{25}$$

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[3] One can *prove* that the Maxwellian distribution (or, in a general Hamiltonian system, the Boltzmann–Gibbs distribution) is the only local equilibrium in the presence of Balescu–Lenard collisions (Dodin, 2022, Appendices E and C.3). But since this can also be anticipated from statistical mechanics, let us skip the formal proof.

[4] In case of a weakly collisional plasma that is mostly gyrotropic with different $P_\perp$ and $P_\parallel$, one can consider similar equations for $P_\perp$ an $P_\parallel$ separately. This leads to a conceptually similar but more complicated model.

where $\boldsymbol{Q} \doteq \int \boldsymbol{w}(mw^2/2) f \,\mathrm{d}\boldsymbol{v}$ is the heat-flux density. Since $\boldsymbol{\epsilon}$ is anti-symmetric and $\boldsymbol{P}$ is symmetric, one has $\epsilon_{ade} P_{ad} = 0$, so the first term on the right-hand side of (25) is actually zero. Also, $\operatorname{tr}(\boldsymbol{P}\nabla\boldsymbol{U}) = (P/D)\nabla\cdot\boldsymbol{U} + \boldsymbol{\Pi} : \nabla\boldsymbol{U}$. This leads to

$$\underbrace{\frac{\partial P}{\partial t} + \boldsymbol{U}\cdot\nabla P}_{\mathrm{d}P/\mathrm{d}t} + \underbrace{\frac{D+2}{D}}_{\gamma} P\,\nabla\cdot\boldsymbol{U} + \frac{2}{D}\nabla\cdot\boldsymbol{Q} + \frac{1}{D}\,\boldsymbol{\Pi} : \nabla\boldsymbol{U} = \frac{2}{D}\,\boldsymbol{S}, \tag{26}$$

where we introduced the collisional heat source $\boldsymbol{S} \doteq \int (mw^2/2)\, C \,\mathrm{d}\boldsymbol{v}$ and γ can be recognized as the polytropic index for adiabatic processes for a monoatomic gas. Also, $\nabla\cdot\boldsymbol{U}$ can be expressed from the continuity equation:

$$\underbrace{\frac{\partial n}{\partial t} + \boldsymbol{U}\cdot\nabla n}_{\mathrm{d}n/\mathrm{d}t} + n\nabla\cdot\boldsymbol{U} = 0, \qquad \nabla\cdot\boldsymbol{U} = -\frac{\mathrm{d}\ln n}{\mathrm{d}t}. \tag{27}$$

Then, one obtains

$$P\,\frac{\mathrm{d}\ln(P/n^\gamma)}{\mathrm{d}t} = \frac{2}{D}\left(\boldsymbol{S} - \nabla\cdot\boldsymbol{Q} - 2\,\boldsymbol{\Pi} : \nabla\boldsymbol{U}\right). \tag{28}$$

This is also commonly expressed as the following equation for the plasma temperature:

$$n\left(\frac{D}{2}\,\frac{\mathrm{d}T}{\mathrm{d}t} + T\nabla\cdot\boldsymbol{U}\right) = \boldsymbol{S} - \nabla\cdot\boldsymbol{Q} - 2\,\boldsymbol{\Pi} : \nabla\boldsymbol{U}, \tag{29}$$

or, equivalently, as

$$P\,\frac{\mathrm{d}}{\mathrm{d}t}\underbrace{\left(\frac{D}{2}\,\ln\frac{P}{n^\gamma}\right)}_{-\sigma} = \boldsymbol{S} - \nabla\cdot\boldsymbol{Q} - 2\,\boldsymbol{\Pi} : \nabla\boldsymbol{U}. \tag{30}$$

Here, σ can be recognized as the per-particle entropy of an ideal monoatomic gas in equilibrium, so (30) can be considered as an entropy equation. Below, we adopt $D = 3$, as usual.

3.3 Chapman–Enskog expansion

Deriving an equation for the scalar pressure and temperature have not solved our problems entirely: we still need equations for $\boldsymbol{S}$, $\boldsymbol{Q}$, $\boldsymbol{\Pi}$, and $\boldsymbol{R}$ too. How can we obtain them? The answer is: perturbation theory. Instead of solving the kinetic equation as is, which would be too complicated, we can search for an approximate solution in the form $f = f_0 + \tilde{f}$, where $\tilde{f} \ll f_0$. We can then find $\tilde{f}$ to the leading order, as a functional of f_0, and calculate the remaining functionals explicitly. This is known as the (generalized) Chapman–Enskog expansion.

Let us start with rewriting the kinetic equation in terms of g defined via $f(t, \boldsymbol{x}, \boldsymbol{v}) = g(t, \boldsymbol{x}, \boldsymbol{v} - \boldsymbol{U})$. Assuming the usual definition $\boldsymbol{w} \doteq \boldsymbol{v} - \boldsymbol{U}$, one finds

$$\partial_t f = \partial_t g - (\partial_t \boldsymbol{U})\cdot\nabla_{\boldsymbol{w}} g, \tag{31a}$$

$$\nabla f = \nabla g - (\nabla\boldsymbol{U})\cdot\nabla_{\boldsymbol{w}} g, \tag{31b}$$

$$\nabla_{\boldsymbol{v}} f = \nabla_{\boldsymbol{w}} g. \tag{31c}$$

This leads to

$$\partial_t g - (\partial_t \boldsymbol{U}) \cdot \nabla_{\boldsymbol{w}} g + (\boldsymbol{U} + \boldsymbol{w}) \cdot \nabla g - (\boldsymbol{U} + \boldsymbol{w}) \cdot (\nabla \boldsymbol{U}) \cdot \nabla_{\boldsymbol{w}} g + \frac{q}{m} \left(\boldsymbol{E} + \frac{\boldsymbol{U}}{c} \times \boldsymbol{B} + \frac{\boldsymbol{w}}{c} \times \boldsymbol{B} \right) \cdot \nabla_{\boldsymbol{w}} g = C, \tag{32}$$

or, equivalently,

$$\partial_t g + (\boldsymbol{U} + \boldsymbol{w}) \cdot \nabla g - \boldsymbol{w} \cdot (\nabla \boldsymbol{U}) \cdot \nabla_{\boldsymbol{w}} g + \left[\frac{q}{m} \left(\boldsymbol{E} + \frac{\boldsymbol{U}}{c} \times \boldsymbol{B} \right) - \frac{\mathrm{d}\boldsymbol{U}}{\mathrm{d}t} \right] \cdot \nabla_{\boldsymbol{w}} g = C - \boldsymbol{w} \times \boldsymbol{\Omega} \cdot \nabla_{\boldsymbol{w}} g, \tag{33}$$

and the expression in the brackets can be further simplified using the momentum equation:

$$\frac{q}{m} \left(\boldsymbol{E} + \frac{\boldsymbol{U}}{c} \times \boldsymbol{B} \right) - \frac{\mathrm{d}\boldsymbol{U}}{\mathrm{d}t} = \frac{\nabla \cdot \boldsymbol{P} - \boldsymbol{R}}{mn} \equiv \boldsymbol{a}. \tag{34}$$

Since collisions between particles of the same type cannot change their own fluid momentum, the friction force $\boldsymbol{R}$ that enters (33) contains contributions only from interactions with *other* species (and the same applies to $\boldsymbol{S}$ in (29) due to energy conservation). For simplicity, let us assume for now that the interaction with other species is weak or negligible. Let us also assume, like in the case of weakly ionized plasma, that ∂_t and ∇ are small too. If we search for a solution in the form $g = g_0 + \tilde{g}$, where g_0 is Maxwellian, then, to the leading order, g can be replaced with g_0 in the entire first line of (33). Also note that the entire second line vanishes at $\tilde{g} \to 0$. Then, (33) can be approximately written as follows:

$$g_0 \hat{\mathcal{D}} \ln g_0 = (\hat{C} - \boldsymbol{w} \times \boldsymbol{\Omega} \cdot \nabla_{\boldsymbol{w}}) \tilde{g} \equiv \hat{\mathcal{C}} \tilde{g}, \tag{35a}$$

$$\hat{\mathcal{D}} \doteq \partial_t + (\boldsymbol{U} + \boldsymbol{w}) \cdot \nabla - \boldsymbol{w} \cdot (\nabla \boldsymbol{U}) \cdot \nabla_{\boldsymbol{w}} + \boldsymbol{a} \cdot \nabla_{\boldsymbol{w}}, \tag{35b}$$

$$\ln g_0 = \ln n - \tfrac{3}{2} \ln T - \tfrac{mw^2}{2T} + \text{const}. \tag{35c}$$

where we used $C[g_0 + \tilde{g}] \approx C_0 + \hat{C}\tilde{g} \approx \hat{C}\tilde{g}$ and $\hat{C}$ is some linear operator.

It now remains to invert the operator $\hat{\mathcal{C}}$. In doing so, one must remember that this operator has a nonzero kernel, so it is not *quite* invertible. However, one can show that $\tilde{g}$ is defined by (35) uniquely under the following constraints (provided that $\hat{C}$ conserves particles, momentum, and energy, as it should):

$$\int \tilde{g} \, \mathrm{d}\boldsymbol{w} = 0, \qquad \int \boldsymbol{w} \tilde{g} \, \mathrm{d}\boldsymbol{w} = 0, \qquad \int w^2 \tilde{g} \, \mathrm{d}\boldsymbol{w} = 0. \tag{36}$$

These constraints mean simply that n, $\boldsymbol{U}$, and $3T/2$ that enter the Maxwellian distribution (24) are actually the true local density, the true local flow velocity, and the true local energy density of the full distribution f. Then, with some perseverance, one can solve (35) explicitly, if $\hat{C}$ is not too complicated. Below, we discuss two examples: a simple toy problem and a complicated but practical result.

3.4 A simple example

Let us assume like-particle collisions, so $\boldsymbol{R} = 0$ and $\boldsymbol{S} = 0$. Then,

$$\begin{aligned}\hat{\mathcal{D}} \ln g_0 &= \left[\frac{\mathrm{d}}{\mathrm{d}t} + \boldsymbol{w}\cdot\nabla + \left(\frac{\nabla P}{mn} - \boldsymbol{w}\cdot\nabla\boldsymbol{U}\right)\cdot\nabla_{\boldsymbol{w}}\right]\ln g_0 \\ &= \left(\frac{\mathrm{d}}{\mathrm{d}t} + \boldsymbol{w}\cdot\nabla\right)\ln n + \left(\frac{w^2}{2v_T^2} - \frac{3}{2}\right)\left(\frac{\mathrm{d}}{\mathrm{d}t} + \boldsymbol{w}\cdot\nabla\right)\ln T + \left(\frac{\nabla P}{mn} - \boldsymbol{w}\cdot\nabla\boldsymbol{U}\right)\cdot\left(-\frac{\boldsymbol{w}}{v_T^2}\right).\end{aligned}$$

From (27) and (29), one has

$$\frac{\mathrm{d}\ln n}{\mathrm{d}t} = -\nabla\cdot\boldsymbol{U}, \qquad \frac{\mathrm{d}\ln T}{\mathrm{d}t} + \frac{2}{3}\nabla\cdot\boldsymbol{U} = 0, \tag{37}$$

where we neglected the effect of $\tilde{g}$, since we are interested only in the leading-order terms. Then,

$$\begin{aligned}\hat{\mathcal{D}}\ln g_0 = -\nabla\cdot\boldsymbol{U} + \boldsymbol{w}\cdot\frac{\nabla n}{n} + \left(\frac{w^2}{2v_T^2} - \frac{3}{2}\right)\left(-\frac{2}{3}\nabla\cdot\boldsymbol{U} + \boldsymbol{w}\cdot\frac{\nabla T}{T}\right) \\ + \left(\frac{T\nabla\ln n}{m} + \frac{\nabla T}{m} - \boldsymbol{w}\cdot\nabla\boldsymbol{U}\right)\cdot\left(-\frac{m\boldsymbol{w}}{T}\right),\end{aligned} \tag{38}$$

or equivalently,

$$\hat{\mathcal{D}}\ln g_0 = \left(\frac{w^2}{2v_T^2} - \frac{5}{2}\right)\boldsymbol{w}\cdot\frac{\nabla T}{T} + \left(\frac{\boldsymbol{w}\boldsymbol{w}}{v_T^2} - \frac{w^2}{3v_T^2}\mathsf{1}\right):\nabla\boldsymbol{U}. \tag{39}$$

Let us also assume $\hat{C} = -\nu$ and $\boldsymbol{B} = 0$, so $\hat{\mathcal{C}} = -\nu$. This leads to

$$\tilde{g} = -\nu^{-1} g_0 \hat{\mathcal{D}}\ln g_0. \tag{40}$$

Using (39) and (40), one finds then, after tedious but straightforward calculations, that

$$\boldsymbol{\Pi} \doteq \int m\left(\boldsymbol{w}\boldsymbol{w} - \mathsf{1}\frac{w^2}{3}\right)\tilde{g}\,\mathrm{d}\boldsymbol{w} = -\frac{P}{\nu}\left[\nabla\boldsymbol{U} + (\nabla\boldsymbol{U})^{\intercal} - \frac{2}{3}(\nabla\cdot\boldsymbol{U})\right], \tag{41}$$

$$\boldsymbol{Q} \doteq \int \frac{mw^2}{2}\boldsymbol{w}\tilde{g}\,\mathrm{d}\boldsymbol{w} = -\frac{5P}{2m\nu}\nabla T. \tag{42}$$

The former contributes as a viscous term to the momentum equation (4):

$$m\left[\frac{\partial\boldsymbol{U}}{\partial t} + (\boldsymbol{U}\cdot\nabla)\boldsymbol{U}\right] = q\left(\boldsymbol{E} + \frac{\boldsymbol{U}}{c}\times\boldsymbol{B}\right) - \frac{\nabla P}{n} + \frac{P}{\nu}\left[\nabla^2\boldsymbol{U} + \frac{1}{3}\nabla(\nabla\cdot\boldsymbol{U})\right] + \ldots$$

Also, $-\nabla\cdot\boldsymbol{Q}$ contributes a diffusive term to (29):

$$\frac{3}{2}\frac{\mathrm{d}T}{\mathrm{d}t} + T\nabla\cdot\boldsymbol{U} = \frac{1}{n}\nabla\cdot\left(\frac{5v_T^2}{2\nu}n\nabla T\right) + \ldots \tag{43}$$

(The remaining term $-2\,\boldsymbol{\Pi}:\nabla\boldsymbol{U}$, which we replaced with ellipsis here, should also be retained under the assumed ordering.) Note that the diffusion coefficient for the temperature is $\sim v_T^2/\nu$.

This is easily understood within the Fokker–Planck paradigm as follows. Collisions between particles of the same type cause order-one change of the particles' individual energies, so one can think of their collisions as collisions of chunks of energy. On a collision time scale $\Delta t \sim \nu^{-1}$, energy propagates a distance $\Delta x \sim \lambda_{\text{mfp}}$. Hence, the diffusion coefficient is supposed to be $\sim (\Delta x)^2/\Delta t \sim \lambda_{\text{mfp}}^2 \nu$, and this is exactly what is predicted by (43). Like in the case of weakly ionized plasma, we can also clearly see now what the small parameter is that we were expanding in: since (43) predicts $\partial_t \sim (\lambda_{\text{mfp}}/L)^2 \nu$, and $\partial_t \ll \nu$ was assumed, the small parameter in this case is the so-called Knudsen number,

$$\text{Kn} \doteq \lambda_{\text{mfp}}/L \ll 1. \tag{44}$$

Also note that, assuming $T_e \sim T_i$, the mean-free paths $\lambda_{ss'} \doteq v_{Ts}/\nu_{ss'}$ of electron and ion collisions satisfy $\lambda_{ee} \sim \lambda_{ei} \sim \lambda_{ii}$, because

$$\nu_{ee} \sim \nu_{ei} \propto m_e^{-1/2}, \qquad \nu_i \sim \nu_{ii} \propto m_i^{-1/2}, \qquad v_{Ts} \propto m_s^{1/2}. \tag{45}$$

This means that, in fully ionized plasmas, Kn is comparable for all species.

3.5 The gold standard: Braginskii equations

A more practical calculation, using the Landau collision operator, was done by Braginskii (1965). The calculation is cumbersome, so we will not discuss it in detail.[5] However, since the Braginskii equations are widely used in practice, let us overview the model and the results. The main assumptions are

$$\frac{\rho_i}{L} \ll \frac{\lambda_{ii}}{L} \sim \sqrt{\frac{m_e}{m_i}} \ll 1 \tag{46}$$

and the "high-flow" ordering, $\boldsymbol{U}_i \sim \boldsymbol{U}_e \sim v_{Ti}$. (An alternative, "low-field" ordering, where the fluid velocities are of order of the diamagnetic drift velocity (15), leads to different results.) Also assumed is $Z_i = 1$ and

$$E_\perp \sim \frac{v_{Ti}}{c} B \sim \frac{L}{\rho_i} \frac{T}{eL} \gg E_\parallel \sim \frac{T}{eL}, \tag{47}$$

$$\frac{\partial}{\partial t} \sim \boldsymbol{U}_s \cdot \nabla \sim \frac{v_{Ti}}{L}, \tag{48}$$

$$U_{i,\parallel} - U_{e,\parallel} \gg \boldsymbol{U}_{i,\perp} - \boldsymbol{U}_{e,\perp} \sim \frac{\rho_i}{L} v_{Ti}. \tag{49}$$

Then, the friction forces scale as

$$R_{ei} \sim R_{ie} \sim n_e m_e \nu_{ei} v_{Ti} \sim \sqrt{\frac{m_e}{m_i}} \frac{L}{\lambda_{ei}} \frac{P_e}{L} \sim \frac{P_e}{L}, \tag{50}$$

[5] A comprehensive tutorial on the Braqinskii equations can be found at https://www-thphys.physics.ox.ac.uk/people/FelixParra/CollisionalPlasmaPhysics/notes/lecIV_braginskii.pdf.

so they can be treated as perturbations. This leads to the following equations for electrons:

$$\nu_{ei} \approx \nu_{ee} = \frac{4\sqrt{2\pi}}{3} \frac{n_e e^4}{m_e^{1/2} T_e^{3/2}} \ln \Lambda_{ee}, \tag{51a}$$

$$\boldsymbol{Q}_e = Q_{e\parallel} \boldsymbol{b} + \boldsymbol{Q}_{e\times} + \boldsymbol{Q}_{e\perp}, \tag{51b}$$

$$Q_{e\parallel} = -\frac{3.16\, P_e}{m_e \nu_{ee}} \boldsymbol{b} \cdot \nabla T_e - 0.71\, P_e (U_{i\parallel} - U_{e\parallel}), \tag{51c}$$

$$\boldsymbol{Q}_{e\times} = -\frac{5 P_e}{2 m_e \Omega_e} \boldsymbol{b} \times \nabla T_e, \tag{51d}$$

$$\boldsymbol{Q}_{e\perp} = -\left(\sqrt{2} + 13/4\right) \frac{P_e \nu_{ee}}{m_e \Omega_e^2} \nabla_\perp T_e - \frac{3 P_e \nu_{ei}}{2\Omega_e} \boldsymbol{b} \times (\boldsymbol{U}_i - \boldsymbol{U}_e), \tag{51e}$$

$$R_{ei,\parallel} = 0.51\, n_e m_e \nu_{ei} (U_{i\parallel} - U_{e\parallel}) - 0.71\, n_e \boldsymbol{b} \cdot \nabla T_e, \tag{51f}$$

$$\boldsymbol{R}_{ei,\perp} = n_e m_e \nu_{ei} (\boldsymbol{U}_{i\perp} - \boldsymbol{U}_{e\perp}) - \frac{3 n_e \nu_{ei}}{2\Omega_e} \boldsymbol{b} \times \nabla T_e, \tag{51g}$$

$$\boldsymbol{S}_{ei} = -\boldsymbol{S}_{ie} + \boldsymbol{R}_{ei} \cdot (\boldsymbol{U}_i - \boldsymbol{U}_e), \tag{51h}$$

and the electron viscosity is negligible, $\boldsymbol{\Pi}_e \approx 0$. For ions, one has

$$\nu_{ii} = \frac{4\sqrt{\pi}}{3} \frac{n_i e^4}{m_i^{1/2} T_i^{3/2}} \ln \Lambda_{ii}, \tag{52a}$$

$$\boldsymbol{Q}_i = Q_{i\parallel} \boldsymbol{b} + \boldsymbol{Q}_{i\times} + \boldsymbol{Q}_{i\perp}, \tag{52b}$$

$$Q_{i\parallel} = -\frac{3.91\, P_i}{m_i \nu_{ii}} \boldsymbol{b} \cdot \nabla T_i, \tag{52c}$$

$$\boldsymbol{Q}_{i\times} = \frac{5 P_i}{2 m_i \Omega_i} \boldsymbol{b} \times \nabla T_i, \tag{52d}$$

$$\boldsymbol{Q}_{i\perp} = -\left(\sqrt{2} + 13/4\right) \frac{P_i \nu_{ii}}{m_i \Omega_i^2} \nabla_\perp T_i, \tag{52e}$$

$$\boldsymbol{R}_{ie} = -\boldsymbol{R}_{ei}, \tag{52f}$$

$$\boldsymbol{S}_{ie} = \frac{3 m_e}{m_i} n_e \nu_{ei} (T_e - T_i), \tag{52g}$$

and the ion viscosity is given by

$$\boldsymbol{\Pi}_i = \Pi_{i\parallel} \left(\boldsymbol{bb} - \tfrac{1}{3}\right) + \boldsymbol{\Pi}_{i\times} + \boldsymbol{\Pi}_{i\perp}, \tag{52h}$$

$$\Pi_{i\parallel} = -\frac{0.96\, P_i}{\nu_{ii}} \left(3\, \boldsymbol{b} \cdot \nabla \boldsymbol{U}_i \cdot \boldsymbol{b} - \nabla \cdot \boldsymbol{U}_i\right), \tag{52i}$$

$$\boldsymbol{\Pi}_{i\times} = \frac{P_i}{4\Omega_i} \left[\boldsymbol{b} \times (\nabla \boldsymbol{U}_i + (\nabla \boldsymbol{U}_i)^\intercal) \cdot (\boldsymbol{1} + 3\boldsymbol{bb}) - (\boldsymbol{1} + 3\boldsymbol{bb}) \cdot (\nabla \boldsymbol{U}_i + (\nabla \boldsymbol{U}_i)^\intercal) \times \boldsymbol{b}\right], \tag{52j}$$

$$\begin{aligned}\boldsymbol{\Pi}_{i\perp} = -\frac{3P_i\nu_{ii}}{10\Omega_i^2}\big[&(\mathbf{1}-\boldsymbol{bb})\cdot(\nabla\boldsymbol{U}_i+(\nabla\boldsymbol{U}_i)^\intercal)\cdot(\mathbf{1}-\boldsymbol{bb})\\&+(\boldsymbol{b}\cdot\nabla\boldsymbol{U}_i\cdot\boldsymbol{b}-\nabla\cdot\boldsymbol{U}_i)(\mathbf{1}-\boldsymbol{bb})\\&+4\big[(\mathbf{1}-\boldsymbol{bb})\cdot(\nabla\boldsymbol{U}_i+(\nabla\boldsymbol{U}_i)^\intercal)\cdot\boldsymbol{bb}+\boldsymbol{bb}\cdot(\nabla\boldsymbol{U}_i+(\nabla\boldsymbol{U}_i)^\intercal)\cdot(\mathbf{1}-\boldsymbol{bb})\big]\big].\end{aligned}\tag{52k}$$

The collision frequencies are the same, up to order-one coefficients, than those we had earlier. The collisional energy exchange between electrons and ions, described by $\boldsymbol{S}_{ie}$, is suppressed by the factor $\sim m_e/m_i$, also as expected. The term $\boldsymbol{R}_{ei}\cdot(\boldsymbol{U}_i-\boldsymbol{U}_e)$ is the Joule heating. It enters the equation for T_e and to for T_i because of the mass ratio: e–i collisions cause electron angular scattering, which heats up electrons but not ions. The fact that $\boldsymbol{R}_{ie}=-\boldsymbol{R}_{ei}$ is simply Newton's third law. The explanation of the coefficients in the heat fluxes is the same as in Sec. 3.4: $\boldsymbol{Q}_{s,\perp}/n_s$ and $Q_{s,\parallel}/n_s$ are the coefficients of the temperature diffusion perpendicular and parallel to $\boldsymbol{B}$, so they scale as $\rho_s^2\nu_s$ and $\lambda_{\mathrm{mfp},s}^2\nu_s$, correspondingly. The fluxes $\boldsymbol{Q}_{s,\times}$ are due to the fact that the temperature gradient makes the average ρ_s space-dependent and thus causes energy flow along $\boldsymbol{b}\times\nabla T_s$, much like ∇B-drift causes particle flow along $\boldsymbol{b}\times\nabla B$.

Since the Braginskii equations assume a short mean-free path, they are inapplicable in the hot plasma core of fusion reactors. Still, they are a standard choice for modeling the cooler, denser edge plasma, where collisionality is high. Also, as the only general analytic collisional closure available, the Braginskii equations commonly serve as a baseline with which other transport models are compared.

Quiz

- What is the particle mobility?
- Derive the conductivity of a weakly ionized plasma.
- What is Fick's flux?
- Derive the diffusion equation for the plasma density for weakly ionized plasmas.
- Estimate the diffusion coefficients for diffusion parallel and perpendicular to magnetic field.
- What is the ambipolar electric field?
- What is ambipolar diffusion?
- Derive the ambipolar diffusion coefficient.
- Estimate the ratio of the electron and ion mobilities for collisions with neutrals.
- Explain the idea of the Chapman–Enskog expansion.
- What is the Knudsen number?
- What is the relation between the mean free paths for e–e, e–i, and i–i collisions?
- Qualitatively, what kind of plasmas can be described by the Braginskii equations?
- Estimate the heat fluxes produced by e–e, e–i, and i–i collisions.

References

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